



GIA - IAA

Introducció a

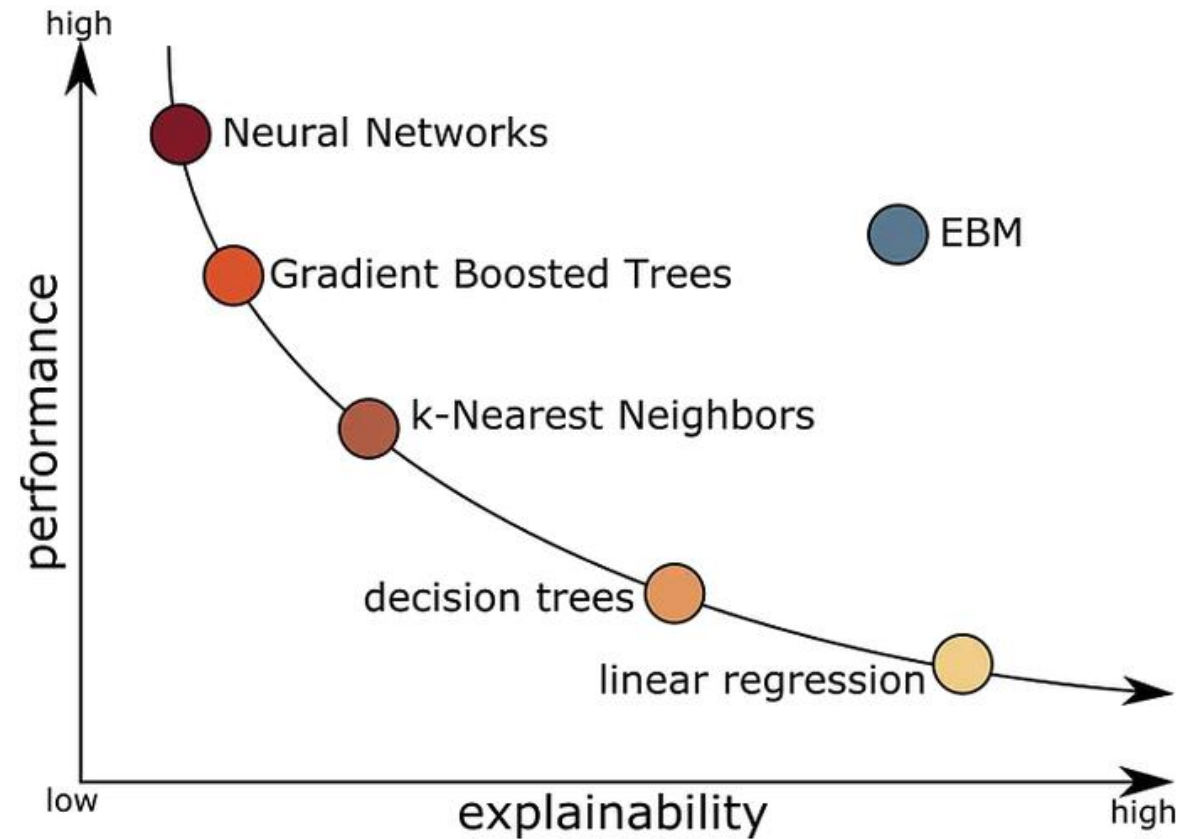
l'Aprenentatge Automàtic

8. Lab: Explainable Boosting Machines

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Do we need explainable models

- ❖ Automated decision-making
- ❖ Even for important decisions
- ❖ ML research: increase accuracy by .01% of an already super complex model
- ❖ Now often extra element: explainability
- ❖ Understudied field of research



InterpretML library

InterpretML

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[build](#) [failing](#) [coverage](#) [82%](#) [maintained](#) [yes](#)

In the beginning machines learned in darkness, and data scientists struggled in the void to explain them. 

Let there be light. 

InterpretML is an open-source package that incorporates state-of-the-art machine learning interpretability techniques under one roof. With this package, you can train interpretable glassbox models and explain blackbox systems. InterpretML helps you understand your model's global behavior, or understand the reasons behind individual predictions.

Interpretability is essential for:

- Model debugging - Why did my model make this mistake?
- Feature Engineering - How can I improve my model?
- Detecting fairness issues - Does my model discriminate?
- Human-AI cooperation - How can I understand and trust the model's decisions?
- Regulatory compliance - Does my model satisfy legal requirements?
- High-risk applications - Healthcare, finance, judicial, ...

```
ebm = ExplainableBoostingClassifier()  
elm.fit(X_train, y_)
```

Train
**Interpretable
Models**

<https://github.com/interpretml/interpret>

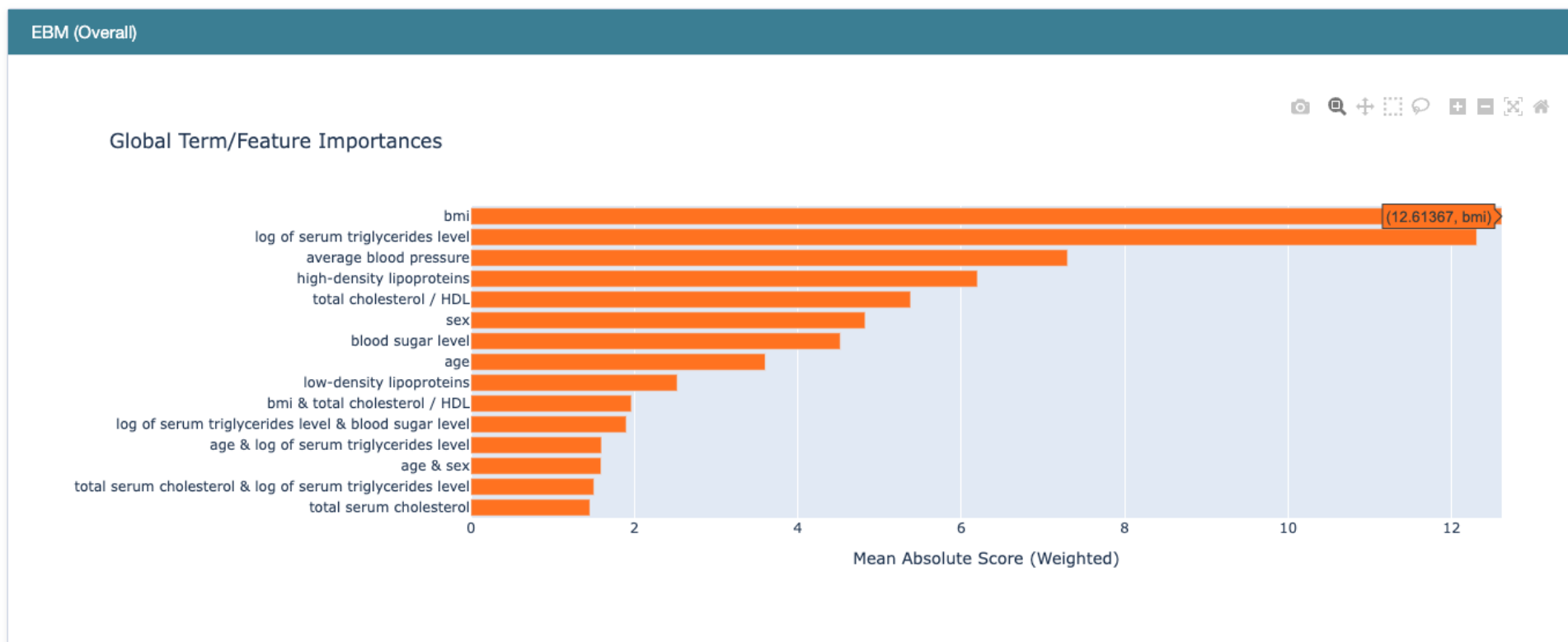
<https://arxiv.org/pdf/1909.09223.pdf>

InterpretML library

```
1 from plotly.offline import iplot
2 ebm_global = ebm.explain_global()
3 show(ebm.explain_global(name='EBM'))
```

[77] ✓ 0.0s

...



<https://github.com/interpretml/interpret>

<https://arxiv.org/pdf/1909.09223.pdf>

Diabetes Dataset

Diabetes Progression

We are going to use a simple regression dataset for explaining this seminar. This dataset uses medical and demographic variables from patients to predict its diabetes progression w.r.t. a baseline.

The diabetes dataset contains measurements taken from 442 diabetic patients:

10 baseline variables (features):

- age – age in years
- sex – male or female
- bmi – body mass index
- bp – average blood pressure
- s1 – TC: total serum cholesterol
- s2 – LDL: low-density lipoproteins
- s3 – HDL: high-density lipoproteins
- s4 – TCH: total cholesterol / HDL
- s5 – LTG: possibly log of serum triglycerides level
- s6 – GLU: blood sugar level

One target variable: a quantitative measure of disease progression one year after baseline

```
1 from sklearn.datasets import load_diabetes
2
3 X, y = load_diabetes(return_X_y=True, scaled=False, as_frame=True)
4 X['sex'] = X['sex'].astype('category')
5 X.sex.value_counts()
```

[4] ✓ 0.0s

```
... sex
1.0    235
2.0    207
Name: count, dtype: int64
```

EBM vs Linear Regression

Linear Regression:

$$E(Y) = \beta_0 + \beta_1.x_1 + \beta_2.x_2 + \dots + \beta_n.x_n$$

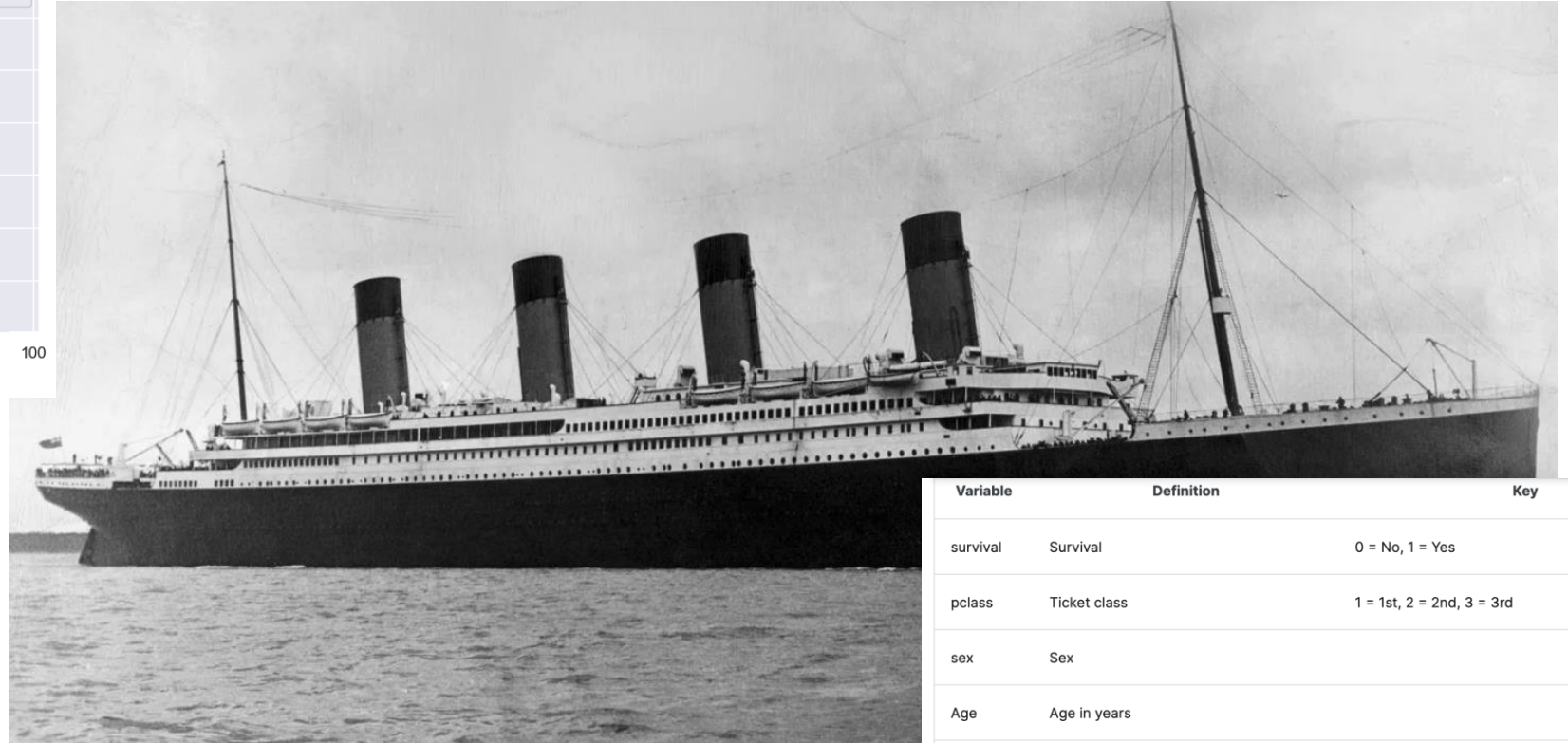
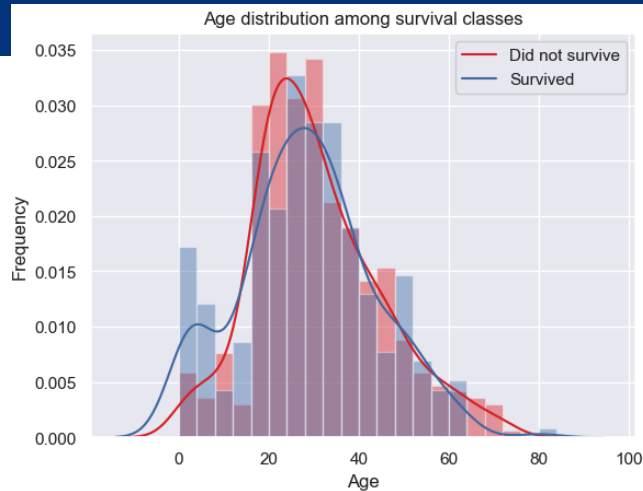
EBM is a type of *Generalized Additive Model (GAM)*, formalized by *Trevor Hastie* and *Robert Tibshirani*:

$$g(E(Y)) = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_n(x_n)$$

GA²M: *Generalized Models with Interactions*:

$$g(E(Y)) = \beta_0 + \sum f_i(x_i) + \sum f_{ij}(x_i, x_j)$$

Titanic Dataset



- ❖ Decide on the task (classification/regression) and train the corresponding EBM to use on the titanic dataset.

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

<https://www.kaggle.com/competitions/titanic/data>